

NATESAN K

Coimbatore, India

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Summary

Computer Science and Engineering graduate with expertise in backend development using Django/DRF and FastAPI, and hands-on experience building AI/ML-integrated full-stack applications. Published IEEE researcher with a proven track record of delivering production-grade platforms and LLM-powered tools, seeking to contribute to innovative full-stack and AI-driven projects in a collaborative environment.

Education

Karpagam Academy of Higher Education

BE Computer Science and Engineering

Coimbatore, India

Sep 2022 – Mar 2026

Experience

Python Developer Intern

OneData Software Solutions

Apr 2026 – Present

Coimbatore, India

- Developing backend services and APIs using Python and modern frameworks.
- Contributing to full-stack development projects with focus on backend optimization and database design.

AI Researcher Intern

National Chung Cheng University

Aug 2025 – Oct 2025

Chiayi, Taiwan

- Contributed to an AI research project focused on bipolar disorder prediction.
- Collected and prepared JSON datasets for analysis using the Reddit API.
- Trained, tuned, and evaluated diverse supervised and unsupervised models to identify the highest-accuracy prediction method.

Freelance Full Stack Developer

Self-Employed

2024 – Present

Coimbatore, India

- Developed and deployed multiple full-stack and AI/ML projects, demonstrating proficiency in backend development and ML integration.

Projects

Disaster Aggregation Software | *GPT-2, Vicuna AI, BLIP-ICM*

- Aggregates disaster-related data from social media and news portals, processes it using LLMs and image captioning models, and presents it on a dashboard for quick response.
- GitHub: BLIP-ICM
- **Published on IEEE Xplore:** ieeexplore.ieee.org/document/11447895

Medical Image Report Generation | *BERT, PyTorch, OpenCV*

- Generates reports for bone fracture X-ray scans using image analysis and NLP models.

Voice-Based Personal Mobile Agent | *Python, React Native, WebSocket, Whisper ASR*

- Developing a real-time AI-powered voice assistant with streaming speech-to-text, intent detection, task automation, cloud-local sync, and modern UI for productivity and task management (Ongoing).
- GitHub: Voice-Agent

References Off | *Next.js, Spring Boot, PostgreSQL, OAuth 2.0*

- Built a full-stack academic collaboration platform (referencesof.com) with Spring Boot 3.4.1 REST API, Next.js frontend, multi-provider OAuth authentication, researcher profiles, social networking, personalized feed, role-based admin panel, and production-grade CI/CD pipeline with Docker, Nginx, and Watchtower.

Hacklyn | *Next.js 16, Django REST Framework, PostgreSQL*

- Developed a full-stack hackathon and event management platform with 40+ API endpoints, multi-role dashboards, OAuth 2.0 authentication, team formation, event lifecycle management, judging system, and a customized Django Admin panel with analytics.
- Live: hacklyn.vercel.app | GitHub: [hacklyn](#)

E-Commerce Web Site | *DRF, Next JS, PostgreSQL, Razorpay, Shiprocket*

- Developed an e-commerce platform for a clothing company with payment integration and shipping management.
- GitHub: Renz-Trending

Academic Course Management System | *DRF, Next JS (TypeScript), Django, React JS*

- Developed a comprehensive platform enabling college students to select courses and access course materials through an LMS interface where teachers can upload course videos and manage student enrollment.
- GitHub: CBCS | EduConnect

Technical Skills

Programming Languages: Python, Java, C/C++, JavaScript

Frameworks & Libraries: Django/DRF, FastAPI, Spring Boot, React JS, React Native, Next JS, Express JS, Axios, Tailwind CSS, LiteLLM, OpenPyXL, Pandas, Numpy

Databases: PostgreSQL, MySQL, SQLite3, MongoDB

AI/ML: PyTorch, OpenCV, NLTK, TensorFlow

Tools & Platforms: Git/GitHub, VS Code, Google Colab, Ngrok, Vercel, AWS, Digital Ocean, Arch Linux, Reddit API

Certifications

AI Using Microcontrollers – CECRI

AI Primer Certification – Infosys

Introduction To GenAI by Google – Coursera

Introduction To Deep Learning – Infosys

Elite Certification In C Programming – NPTEL

Awards

HackTheFuture (Earth Edition): Secured 3rd place in Edu Tech theme (LMS with GenAI)

Dell-AI-thon: Secured 3rd place for an innovative AI solution

TechSonance'23, PSGCAS: Won 3rd prize in a problem-solving Event

HIFOCUS'25, HIT: Won 1st place in Code Debugging Event